

Database Management Systems

CPS 3740

Lecture 11

Computer Science, Kean University

Dr. Ching-yu (Austin) Huang

Instructor Paolien Wang

Outline

- MySQL Workbench
- SQL DML - INSERT, UPDATE, DELETE

MySQL Workbench

- MySQL Workbench is a Graphic User Interface (GUI) tool to interact with MySQL database
 - A client, NOT a database server.
- Install
- Setup connection
- SQL tab
- Features



MySQL – Workbench download/setup

- Download client at
 - <https://www.mysql.com/products/workbench/>
- Configuration
 - Connection name: **your own label name**
 - Hostname: **imc.kean.edu**
 - Port: **3306**
 - Username: **Your login and password**
 - Default schema: **class database (CPS3740_20xxxx)**
- Test connection and OK
- Document
 - <https://dev.mysql.com/doc/workbench/en/wb-what-is-new-62.html>

MySQL Workbench configuration

Connection Name: **CPS3740 xxxx**

Connection Remote Management System Profile

Connection Method: Standard (TCP/IP) Method to use to connect to the RDBMS

Parameters SSL Advanced

Hostname: imc.kean.edu Port: 3306 Name or IP address of the server host. - TCP/IP port.

Username: **XXXX** Name of the user to connect with.

Password: Store in Vault ... Clear The user's password. Will be requested later if it's not set.

Default Schema: **Class database** The schema to use as default schema. Leave blank to select it

Duplicate Move Up Move Down Test Connection Close

Hostname:
imc.kean.edu

Workbench GUI

The screenshot displays the MySQL Workbench interface. The top menu bar includes File, Edit, View, Query, Database, Server, Tools, Scripting, and Help. The left sidebar contains a 'MANAGEMENT' section with links to Server Status, Client Connections, Users and Privileges, Status and System Variables, Data Export, and Data Import/Restore. Below this is the 'INSTANCE' section with Startup / Shutdown, Server Logs, and Options File. The 'SCHEMAS' section shows a tree view of databases, with 'actor' selected. The main query editor shows a query: `SELECT * FROM sakila.actor;`. The results pane displays a table with 10 rows of actor data. The bottom pane shows the 'Output' tab with a message: '200 row(s) returned'.

Table: actor

actor_id	first_name	last_name	last_update
1	PENELOPE	GUINNESS	2006-02-15 04:34:33
2	NICK	WAHLBERG	2006-02-15 04:34:33
3	ED	CHASE	2006-02-15 04:34:33
4	JENNIFER	DAVIS	2006-02-15 04:34:33
5	JOHNNY	LOLLOBRIGIDA	2006-02-15 04:34:33
6	BETTE	NICHOLSON	2006-02-15 04:34:33
7	GRACE	MOSTEL	2006-02-15 04:34:33
8	MATTHEW	JOHANSSON	2006-02-15 04:34:33
9	JOE	SWANK	2006-02-15 04:34:33
10	CHRISTIAN	GABLE	2006-02-15 04:34:33

Output

Time	Action	Message	Duration / Fetch
1 18:58:01	SELECT * FROM sakila.actor LIMIT 0, 1000	200 row(s) returned	0.016 sec / 0.000 sec

Work on the Workbench

- Test and practice Workbench
 - List all columns and rows in the Staff table.
> `select * from dreamhome.Staff;`
 - List the number of staff and the average salary for each branch
> `??`
 - List the monthly salary for every staff
> `??`
- Tutorial
 - https://www.youtube.com/watch?v=X_umYKqKaF0

See the Staff table structure

- Use the command DESC

```
mysql> desc Staff;
```

Field	Type	Null	Key	Default	Extra
staffNo	varchar(5)	NO	PRI	NULL	
fName	varchar(15)	NO		NULL	
lName	varchar(15)	NO		NULL	
position	varchar(25)	YES		NULL	
sex	char(1)	YES		NULL	
DOB	date	YES		NULL	
salary	decimal(8,2)	YES		NULL	
branchNo	varchar(8)	YES		NULL	

Class exercise – create your own Staff table

- Use your own Staff table in class database CPS3740_XXXX (eg: CPS3740_2023S)
- Create Staff_XXXX table (no foreign key)

```
Create table Staff_XXXX (  
  staffNo varchar(5) NOT NULL,  
  fName varchar(15) NOT NULL,  
  lName varchar(15) NOT NULL,  
  position varchar(25) DEFAULT NULL,  
  sex char(1) DEFAULT NULL,  
  DOB date DEFAULT NULL,  
  salary decimal(8,2) DEFAULT NULL,  
  branchNo varchar(8) DEFAULT NULL,  
  PRIMARY KEY (staffNo));
```

- Get insert, update, delete privileges on the class website.
- Discuss more about primary key and create tables later.

INSERT

INSERT INTO TableName [(columnList)]
VALUES (dataValueList)

- *columnList* is optional; if omitted, SQL assumes a list of all columns in their original CREATE TABLE order.
- Any columns omitted must have been declared as NULL when table was created, unless DEFAULT was specified when creating column.

Rules for the INSERT

- *dataValueList* must match *columnList* as follows:
 - **number of items** in each list must be same;
 - must be direct correspondence in position of items in two lists;
 - data type of each item in *dataValueList* must be compatible with **data type** of corresponding column.

Example 6.35 INSERT ... VALUES

- Insert a new row into Staff table supplying data for all columns.

INSERT INTO Staff VALUES ('SG16', 'Alan', 'Brown', 'Assistant', 'M', Date '1957-05-25', 8300, 'B003');

```
mysql> desc Staff;
```

Field	Type	Null	Key	Default	Extra
staffNo	varchar(5)	NO	PRI	NULL	
fName	varchar(15)	NO		NULL	
lName	varchar(15)	NO		NULL	
position	varchar(25)	YES		NULL	
sex	char(1)	YES		NULL	
DOB	date	YES		NULL	
salary	decimal(8,2)	YES		NULL	
branchNo	varchar(8)	YES		NULL	

Example 6.36 INSERT using Defaults 1

- Insert a new row into Staff table supplying data for all mandatory columns, but not all values.

```
INSERT INTO Staff (staffNo, fName, lName,  
                  position, salary, branchNo)  
VALUES ('SG44', 'Anne', 'Jones',  
        'Assistant', 8100, 'B003');
```

- Check what happens to columns – sex, salary, DOB.
 - The values for these unknown will be depended on the default values.
 - Check what the default values are for these 3 columns

Example 6.36 INSERT using Defaults 2

- Insert a new row into Staff table with some unknown values.
- Without column names, we still need to provide all values

INSERT INTO Staff

**VALUES ('SG44', 'Anne', 'Jones', 'Assistant', NULL,
NULL, 8100, 'B003');**

- Do it on your own Staff_xxxx table

Class exercise - Insert

- What happens to the following INSERT statements?
 1. **INSERT INTO Staff (staffNo, fName, lName, position, salary, branchNo)**
VALUES ('S123456', 'Anne', 'Jones', 'Assistant', 8100, 'B003');
 2. **INSERT INTO Staff (staffNo, fName, lName, position, salary, branchNo)**
VALUES ('S12', , 'Jones', , 8100, 'B003');
 3. **INSERT INTO Staff (staffNo, fName, lName, salary)**
VALUES ('S123456', 'Anne', 'Jones', 'Assistant', 'B003'); --check values
 4. **INSERT INTO Staff (staffNo, fName, branchNo)**
VALUES ('S123', 'Anne', 8100, 'B003');
 5. **INSERT INTO Staff (staffNo, fName, branchNo) VALUES ('S123', 8100, 'B003');**
 6. **INSERT INTO Staff (fName, lName, salary) VALUES ('Anne', 8100, 'Jones');**
- **Project tip – debug:** print your PHP INSERT query on the browser, and copy/paste to run the workbench or commandline

Class exercise - Insert More Than One Row

INSERT INTO Staff VALUES

```
('SG44', 'Anne1', 'Jones', 'Assistant', NULL, NULL, 8100, 'B003'),  
('SG45', 'Anne2', 'Jones', 'Assistant', NULL, NULL, 8100, 'B003'),  
('SG46', 'Anne3', 'Jones', 'Assistant', NULL, NULL, 8100, 'B003');
```

- Do it on your own Staff_xxxx table

Class exercise - Insert special characters

- Do it on your own Staff_xxxx table
- How to insert the record if someone's last name is "O'Hello"?

Insert into Staff_xxxx (staffno, lname) values
('ab12', 'O'hello'); → illegal, syntax error

INSERT ... SELECT

- How do you copy a lot of records from one table to another table?
 - We don't want to manually insert them!
- Second form of INSERT allows multiple rows to be copied from one or more tables to another:

```
INSERT INTO TableName [ (columnList) ]  
    SELECT ...
```

Class exercise – INSERT ... SELECT

- Create your own staff table alled **Staff_xxxx** in database **CPS3740_2020F** on **imc.kean.edu**
- How do we copy the existing records from Staff to your staff table?

```
mysql> INSERT into Staff_xxxx  
        SELECT * from dreamhome.Staff;
```

Example 6.37 INSERT ... SELECT

- Assume there is a table StaffPropCount that contains names of staff and number of properties they manage:

StaffPropCount(staffNo, fName, lName, propCnt)

- Populate StaffPropCount using Staff and PropertyForRent tables.

<Answer> next slide

Example 6.37 INSERT ... SELECT

```
INSERT INTO StaffPropCount
(SELECT s.staffNo, fName, lName, COUNT(*)
FROM Staff s, PropertyForRent p
WHERE s.staffNo = p.staffNo
GROUP BY s.staffNo, fName, lName)
UNION
(SELECT staffNo, fName, lName, 0
FROM Staff
WHERE staffNo NOT IN
  (SELECT DISTINCT staffNo
   FROM PropertyForRent));
```

Example 6.37 INSERT ... SELECT

staffNo	fName	lName	propCount
SG14	David	Ford	1
SL21	John	White	0
SG37	Ann	Beech	2
SA9	Mary	Howe	1
SG5	Susan	Brand	0
SL41	Julie	Lee	1

- If second part of UNION is omitted, excludes those staff who currently do not manage any properties.

Project: MySQL in PHP - INSERT

```
Include "dbconfig.php";
$con = mysqli_connect($host, $user, $pass , $dbname);
$query = " INSERT INTO ..... " ;
echo "<br>$query\n";
$result = mysqli_query($con, $query);
# how do you the record is inserted successfully?
if($result) {
    echo "Insert ... successfully!";
}
else
    echo " Something wrong ... ";
# mysqli_free_result($result); No result set return!!!!
mysqli_close($con);
```

UPDATE

UPDATE TableName

SET columnName1 = dataValue1

[, columnName2 = dataValue2...]

[**WHERE** searchCondition]

- *TableName* can be name of a base table or an updatable view.
- SET clause specifies names of one or more columns that are to be updated.

UPDATE

- **WHERE clause is optional:**
 - if omitted, named columns are updated for all rows in table;
 - if specified, only those rows that satisfy *searchCondition* are updated.
- New *dataValue(s)* must be compatible with data type for corresponding column.

Example 6.38/39 UPDATE All Rows

Give all staff a 3% pay increase.

UPDATE Staff

SET salary = salary*1.03;

Give all Managers a 5% pay increase.

UPDATE Staff

SET salary = salary*1.05

WHERE position = 'Manager';

Example 6.40 UPDATE Multiple Columns

Promote David Ford (staffNo='SG14') to Manager and change his salary to £18,000.

UPDATE Staff

SET position = 'Manager', salary = 18000

WHERE staffNo = 'SG14';

Class exercise - update

- Students spend 5 minutes to do these questions on their own Staff_xxxx table
- Increase 10% salary for the female whose position is secretary.
- Insert a record with your own information into your own Staff_xxxx table:
 - Position is “CXO”
 - Salary = -100000
- Fix your salary to 100000 and position to “CEO”

Class exercise - Update special characters

- Do it on your own Staff_xxxx table
- Change someone's last name to "O'Hello"
- Change the position "Assistant" to "Associate, II"

DELETE

DELETE FROM TableName
[WHERE searchCondition]

- *TableName* can be name of a base table or an updatable view.
- *searchCondition* is optional; if omitted, all rows are deleted from table. This does not delete table. If *search_condition* is specified, only those rows that satisfy condition are deleted.

Example 6.41/42 DELETE Specific Rows

- Delete all viewings that relate to property PG4.

**DELETE FROM Viewing
WHERE propertyNo = 'PG4';**

- Delete all records from the Viewing table.

DELETE FROM Viewing;

- Do it on your own Staff_xxxx table
 - Delete Mary from your Staff_xxxx table

Class exercise –delete & create/copy

- Write a SQL query to delete your record from your own Staff_xxxx table.
- Create your own Staff_xxxx and copy all the records from dreamhome.Staff at the same time
`Create table Staff_xxxx as select * from dreamhome.Staff;`

Project

- Check status
- Help students' problems
- Seek help from Samurai as earlier as possible

Conclusions

- Install and familiar with MySQL Workbench
- SQL exercise questions 78 - 87
- Try the examples in textbook/slides
 - INSERT
 - UPDATE
 - DELETE